

Question 1. (2 marks) Let $\{X_n\}_{n=0}^\infty$ be a Markov Chain on the state space $S = \{1, 2, 3\}$ and with the T.P. matrix $P = (p_{ij})_{i,j \in S}$ given by

$$P = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \end{pmatrix}.$$

Given that $\mathbb{P}(X_0 = i) = \frac{1}{3}, \forall i \in S$. Compute $\mathbb{P}(X_0 = i | X_1 = 2)$ for all $i \in S$.

Hints: We have

$$\mathbb{P}(X_0 = i | X_1 = 2) = \frac{\mathbb{P}(X_0 = i, X_1 = 2)}{\mathbb{P}(X_1 = 2)} = \frac{\mathbb{P}(X_0 = i) \times p_{i2}}{\sum_{j=1}^3 \mathbb{P}(X_0 = j) \times p_{j2}} = \frac{p_{i2}}{\sum_{j=1}^3 p_{j2}}$$

Question 2. Prove the following statements.

- (a) (2 marks) Let C be any communicating class. Then no proper subset of C is closed.
- (b) (3 marks) Suppose i is transient and $j \rightarrow i$. Then $\sum_{n=1}^\infty p_{ij}^n < \infty$.
- (c) (3 marks) A Markov Chain on a finite state space has at least one recurrent state.
- (d) (2 marks) Consider a Markov chain $\{X_n\}_{n \geq 0}$ on the state space $\{0, 1, 2, 3\}$ and the T.P. matrix

$$\begin{pmatrix} \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 \\ 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} \end{pmatrix}.$$

Then the state 3 is transient.

Hints:

- (a) Let A be a proper subset of C . Choose $i \in A$ and $j \in C \setminus A$. Since $i, j \in C$ and C is a communicating class, $i \leftrightarrow j$. In particular, $i \in A$ and $i \rightarrow j$ with $j \notin A$.
- (b) As $j \rightarrow i$, there exists a natural number m with $p_{ji}^m > 0$. By Chapman-Kolmogorov relations and since i is transient, we have

$$\left(\sum_{n=1}^\infty p_{ij}^n \right) \times \underbrace{p_{ji}^m}_{>0} \leq \sum_{n=1}^\infty p_{ii}^{n+m} < \infty$$

- (c) Proof by contradiction. If possible, assume all the states are transient. Then for all i, j in the state space S , $\lim_{n \rightarrow \infty} p_{ij}^n = 0$. Get a contradiction using $\sum_{j \in S} p_{ij}^n = 1, \forall i \in S$ and that S is finite.
- (d) For $i = 0, 1, 2$, $p_{i3} = (P)_{i3} = 0$. For $n \geq 2$,

$$p_{i3}^n = (P^n)_{i3} = (P^{n-1} \times P)_{i3} = \sum_{k=0}^3 (P^{n-1})_{ik} \times (P)_{k3} = (P^{n-1})_{i3} \times p_{33} = 0,$$

by induction. Hence, $i \nrightarrow 3$ for $i = 0, 1, 2$. Consequently, $f_{33}^1 = p_{33} = \frac{1}{2}$ and $f_{33}^n = 0, n \geq 2$.

Since $f_{33} = \sum_{n=1}^\infty f_{33}^n < 1$, the state 3 is transient.